



ABHISHEK GOYAL



ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
---	M.Sc in Mathematics	Indian Institute of Technology Delhi	7.8
2025	B.Sc - B.Ed (Integrated)	University Of Rajasthan	72.98%
2021	CBSE	SR International Academy Nadbai	88.2%
2019	CBSE	SR International Academy Nadbai	90.2%

IIT DELHI THESIS

Title: A Comparative Analysis of Supervised Classification Methods under **Prof. Harminder Ghalwat** (Aug'26 – Present)

Description: Investigating mathematical foundations, modelling assumptions, calibration behaviour, and practical tradeoffs of classical supervised classifiers across diverse real-world dataset conditions and scenarios

- Implementing major classifiers from scratch in NumPy/SciPy and validating against scikit-learn implementations.
- Benchmarking performance, calibration, efficiency, and robustness under noise, outliers, and varying regimes.

PROJECTS

• CivicStruct - Small-LLM QLoRA Fine-Tuning, Reproducible & Rigorous Evaluation (Jul'26 - Aug'26)

- Built an NLP system converting informal Indian civic complaints into **schema-validated JSON** (domain, urgency, missing-info, factual fields) against a 7-domain / 8-issue taxonomy enforced by dependency-free Python validator.
- Ran **3-model bake-off** (Qwen3-4B, SmoLLM3-3B, Phi-4-mini) across zero/few/retrieved-shot; selected **SmoLLM3-3B** on schema/missing-info F1; 40/80/160-row x 3-seed QLoRA ablations raised validation domain F1 from 0.703 to 0.904.
- Fine-tuned **4-bit QLoRA adapter (30.2M params)** to **0.94 schema validity / 0.829 domain F1 / 0.745 issue F1** on 50 held-out complaints with 95% CIs; added **MLflow registry gates**, DVC-locked evaluation, CI, FastAPI, Docker & Gradio.

• Tech Salary Advisor - End-to-End MLOps & Disciplined Regression Pipeline (Jun'26 - Aug'26)

- Engineered a modular scikit-learn pipeline on ~110k Indian tech salary profiles with train-fitted imputation, log1p target transform, raw-data validation, lineage tracking, and **four disjoint train/validation/calibration/test splits**.
- Packaged inference as a **Dockerized FastAPI** service with Streamlit UI and calibrated **90% split-conformal salary intervals** (90.4% held-out coverage); GitHub Actions runs Ruff, **33 tests**, builds, and smoke-tests /health and /predict.
- Built a **champion/challenger MLflow Registry** with validation-gated promotion (MAE <+2%, R2 >-0.005); Optuna-tuned CatBoost beat **8 baselines** (incl. a Keras ANN), scoring **R2 0.884 / 1.55 LPA MAE** on 21k untouched test rows.

• Signature Forgery Verification - Metric Learning, Leak Detection & Reproducible Gate (May'26 - Jul'26)

- Built an offline signature-verification system across Latin & Devanagari scripts (**224 writers**) that verifies unseen writers without retraining, implementing a **writer-disjoint split** protocol and a leak-free hard-pair sampling pipeline.
- Identified **2 severe dataset-leakage bugs** in public datasets (duplicate test splits + folder shortcuts) and automated an md5-based check-data-leak.py validation gate that exits non-zero on compromised data before any training run.
- Trained twin EfficientNet-B0 towers with batch-hard triplet loss to reach **0.986 ROC-AUC / 93.9% accuracy** on 51 unseen writers (**0.877 NFI AUC**); packaged and served a 17 MB model on HF Spaces via GitHub Actions CI/CD.

• Dark Pattern Text Risk Screener - Multi-Model NLP Pipeline & Dual Deployment (May'26 - Jul'26)

- Engineered a **14-class NLP** screening pipeline on **6,373** e-commerce samples aligned to India's 2023 CCPA dark-pattern taxonomy, using source-page grouping to prevent cross-validation leakage for cleaner evaluation.
- Fine-tuned **DistilBERT** to **0.883 macro-F1 / 91.1% accuracy** and built a **4.4 MB pipeline** (char TF-IDF + 12 features + SMOTE + calibrated LinearSVC); deployed on **Streamlit & HF Spaces** with 0.50-threshold abstention gating.

TECHNICAL SKILLS

- **Programming & ML/NLP:** Python, SQL | PyTorch, TensorFlow/Keras, NumPy, SciPy, HF Transformers, PEFT/QLoRA
- **Classical ML & Evaluation:** Scikit-learn, CatBoost, Optuna, SMOTE, Permutation Importance, Bootstrap CIs
- **MLOps & Deployment:** MLflow, DVC, Docker, FastAPI, Streamlit, Gradio, Pytest, GitHub Actions, CI/CD
- **Links:** [GitHub](#), [Portfolio](#) | **Certifications:**
 - **DL Specialization** - DeepLearning.AI: ANN, CNNs, RNNs, LSTMs, Sequence Models, Attention & Transformers
 - **ML Engineer Track** - DataCamp : MLOps, MLflow, Docker, Data Pipelines, CI/CD, Model Deployment & Monitoring

EXTRA CURRICULAR ACTIVITIES

- Cricket enthusiast and recreational hiker, actively participating in outdoor and team-based activities.
- Completed a **4-month Mathematics teaching internship**, delivering classroom instruction and assessments.
- Completed **British Airways Data Science Job Simulation**, analysing reviews and modelling buying behaviour.



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IIT COURSE

Degree	Institute	CGPA	Dept. Rank
M.Sc in Mathematics	Indian Institute of Technology Delhi	7.8	16

COURSES DONE

Stochastic Processes And Its Applications, Functional Analysis, Statistical Inference, Mathematical Programming, Introduction To Computer Programming, Introduction To Probability, Real Analysis, Linear Algebra